

Pandas (Series & DataFrame) in Python - Complete Guide

What is Pandas?

Pandas is a powerful Python library used for:

- Data Analysis
- Data Cleaning
- Data Manipulation
- Data Visualization Preparation
- Working with CSV, Excel, SQL Data

Installation

```
pip install pandas
```

Import Pandas

```
import pandas as pd
```

1. Pandas Series

A **Series** is a one-dimensional labeled array.

Creating a Series

```
import pandas as pd

s = pd.Series([10, 20, 30, 40, 50])

print(s)
```

Output:

```
0    10
1    20
2    30
3    40
4    50
dtype: int64
```

Series with Custom Index

```
s = pd.Series([10, 20, 30], index=['a', 'b', 'c'])
```

```
print(s)
```

Output:

```
a    10
b    20
c    30
```

Access Elements

By Index

```
print(s['a'])
```

Output:

```
10
```

By Position

```
print(s.iloc[0])
```

Output:

```
10
```

Series Attributes

```
print(s.index)
print(s.values)
print(s.dtype)
print(s.shape)
print(s.size)
```

Series Methods

Sum

```
s.sum()
```

Mean

```
s.mean()
```

Maximum

```
s.max()
```

Minimum

```
s.min()
```

Count

```
s.count()
```

Series Operations

```
s = pd.Series([10, 20, 30])

print(s + 5)
print(s * 2)
```

Output:

```
0    15
1    25
2    35
```

Series from Dictionary

```
data = {
    'A': 10,
    'B': 20,
    'C': 30
}

s = pd.Series(data)

print(s)
```

Output:

```
A    10
B    20
C    30
```

2. Pandas DataFrame

A **DataFrame** is a two-dimensional table.

Similar to an Excel Sheet.

Name	Age	City
Krish	20	Rajkot
Amit	21	Surat

Name	Age	City
Ravi	22	Ahmedabad

Creating DataFrame

From Dictionary

```
data = {  
    "Name": ["Krish", "Amit", "Ravi"],  
    "Age": [20, 21, 22],  
    "City": ["Rajkot", "Surat", "Ahmedabad"]  
}  
  
df = pd.DataFrame(data)  
  
print(df)
```

Output:

	Name	Age	City
0	Krish	20	Rajkot
1	Amit	21	Surat
2	Ravi	22	Ahmedabad

DataFrame Attributes

```
print(df.shape)  
print(df.columns)  
print(df.index)  
print(df.dtypes)  
print(df.size)  
print(df.ndim)
```

DataFrame Information

info()

```
df.info()
```

Output:

```
<class 'pandas.core.frame.DataFrame'>  
RangeIndex: 3 entries
```

describe()

```
df.describe()
```

Provides:

- Count
- Mean
- Std
- Min
- Max

View Data

First 5 Rows

```
df.head()
```

First 2 Rows

```
df.head(2)
```

Last Rows

```
df.tail()
```

Selecting Data

Single Column

```
df["Name"]
```

Output:

```
0    Krish
1     Amit
2     Ravi
```

Returns a **Series**

Multiple Columns

```
df[["Name", "Age"]]
```

Returns a **DataFrame**

Row Selection

loc[] (Label Based)

```
df.loc[0]
```

Output:

Name	Krish
Age	20
City	Rajkot

iloc[] (Position Based)

```
df.iloc[0]
```

Multiple Rows

```
df.iloc[0:2]
```

Output:

Krish
Amit

Adding Columns

```
df["Salary"] = [30000, 40000, 50000]  
  
print(df)
```

Updating Data

```
df.loc[0, "Age"] = 25
```

Deleting Data

Drop Column

```
df.drop("Salary", axis=1)
```

Permanent

```
df.drop("Salary", axis=1, inplace=True)
```

Drop Row

```
df.drop(0)
```

Filtering Data

Age > 20

```
df[df["Age"] > 20]
```

Output:

	Name	Age	City
1	Amit	21	Surat
2	Ravi	22	Ahmedabad

Multiple Conditions

```
df[(df["Age"] > 20) & (df["City"] == "Surat")]
```

Sorting

Sort by Age

```
df.sort_values("Age")
```

Descending:

```
df.sort_values("Age", ascending=False)
```

Missing Values

Check Missing Values

```
df.isnull()
```

Count Missing Values

```
df.isnull().sum()
```

Fill Missing Values

```
df.fillna(0)
df.fillna(df["Age"].mean())
```

Remove Missing Values

```
df.dropna()
```

Renaming Columns

```
df.rename(columns={"Name": "Student_Name"})
```

Index Operations

Set Index

```
df.set_index("Name")
```

Reset Index

```
df.reset_index()
```

GroupBy

```
data = {
    "Department": ["IT", "IT", "HR", "HR"],
    "Salary": [50000, 60000, 45000, 55000]
}
```

```
df = pd.DataFrame(data)
```

```
df.groupby("Department")["Salary"].mean()
```

Output:

```
Department
```

```
HR      50000
IT      55000
```

Reading Files

CSV File

```
df = pd.read_csv("data.csv")
```

Excel File

```
df = pd.read_excel("data.xlsx")
```

Writing Files

CSV

```
df.to_csv("output.csv", index=False)
```

Excel

```
df.to_excel("output.xlsx", index=False)
```

Series vs DataFrame

Feature	Series	DataFrame
Dimension	1D	2D
Rows	Yes	Yes
Columns	No	Yes
Single Data Type	Usually	Multiple
Example	Marks	Student Table

Interview Questions

1. What is Pandas?

Python library for data analysis and manipulation.

2. Difference between loc and iloc?

loc	iloc
Label Based	Position Based
Uses Index Name	Uses Integer Position

3. Difference between Series and DataFrame?

Series = 1D

DataFrame = 2D

4. What is GroupBy?

Used to split data into groups and perform operations.

5. What is NaN?

Missing value in Pandas.

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